

CS173: Discrete Structures

Problem Set 6

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1. You may assume all theorems up to and including the Fundamental Theorem of Arithmetic, Euclid's Division Lemma, Bézout's Identity, and Euclid's Lemma.

Use induction to prove that $(\forall n \in \mathbb{N})(5 \mid 8^n - 3^n)$.

Proof. We proceed by weak induction on n . **Base case** ($n = 0$): We have $8^0 - 3^0 = 1 - 1 = 0 = 5 \cdot 0$, so $5 \mid 8^0 - 3^0$. ✓

Inductive hypothesis: Let $k \in \mathbb{N}$ and assume $5 \mid 8^k - 3^k$. By definition of divisibility, there exists $c \in \mathbb{Z}$ such that $8^k - 3^k = 5c$.

Inductive step: We need to show $5 \mid 8^{k+1} - 3^{k+1}$. Observe that:

$$\begin{aligned} 8^{k+1} - 3^{k+1} &= 8 \cdot 8^k - 3 \cdot 3^k \\ &= 8 \cdot 8^k - 8 \cdot 3^k + 8 \cdot 3^k - 3 \cdot 3^k \\ &= 8(8^k - 3^k) + 5 \cdot 3^k \\ &= 8(5c) + 5 \cdot 3^k \\ &= 5(8c + 3^k) \end{aligned}$$

Since $8c + 3^k \in \mathbb{Z}$, we have $5 \mid 8^{k+1} - 3^{k+1}$. ✓

Therefore by weak induction, $(\forall n \in \mathbb{N})(5 \mid 8^n - 3^n)$. □